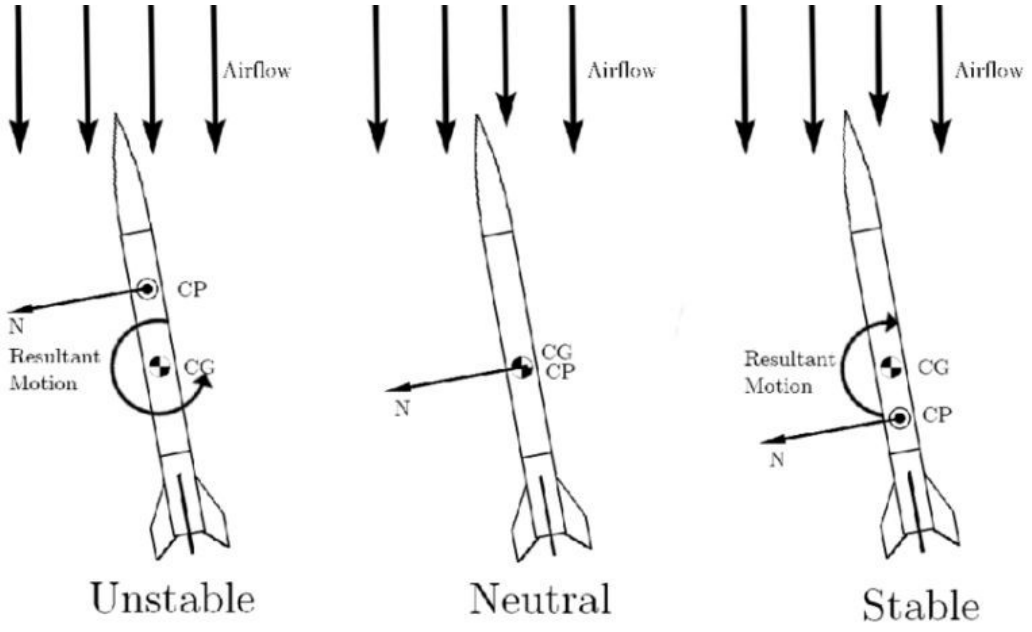
Stability explaining

(figure from https://www.linkedin.com/posts/aurangjeb22_aerospaceengineering-rocketscience-aerodynamics-activity-7399389787914326016-kRmT)



Estes Vapor Model Rocket parameters:

Body tube diameter: 1.63 inches (41.4 mm)

Empty Weight: 3.5 oz (99.22 g)

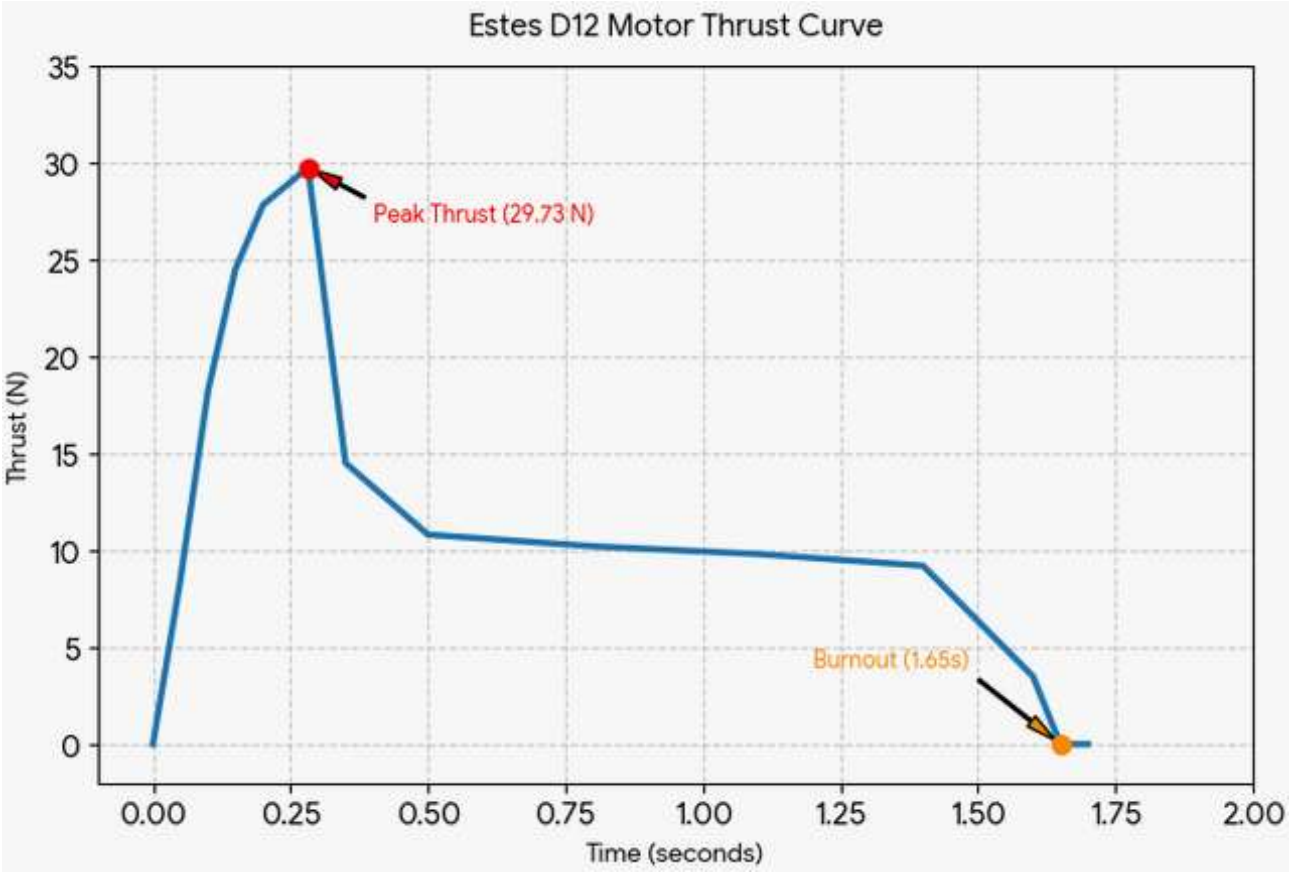


D12-5 Engine data:

Total Mass: 1.61 oz (45.6 g)

Propellant Mass: 0.85 oz (24.2 g)

Thrust data (curve):



Simulate fly !

Setup a time step dt for each step

1 Evaluate total force of the rocket

Total force = + Thrust (interpolate from the above curve)

- Drag force (from Cd and live speed)
- Weight

2 derive acceleration, Newton's 2nd law, F=ma.

(total mass lost assuming propellant burns off linearly over engine time)

3 assume this instantaneous acceleration is a constant, then current velocity is

$V_t = a_{t-1} + a_t * dt$

4 current step altitude climbing:

$H_t = V_t * dt$

5 Total height updates

Loop to Next time step